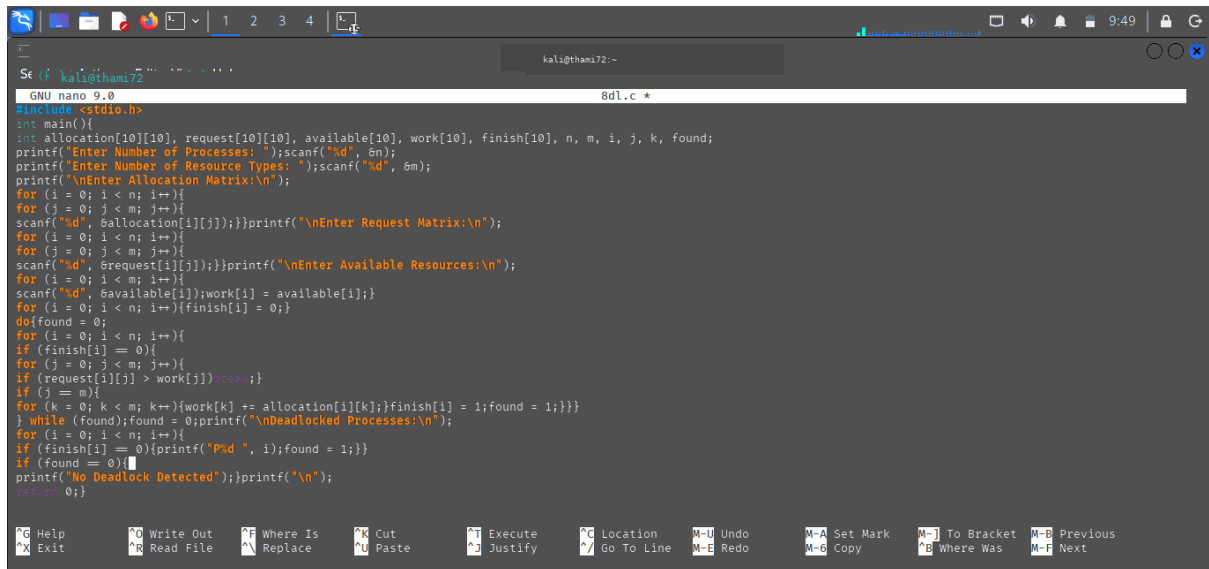
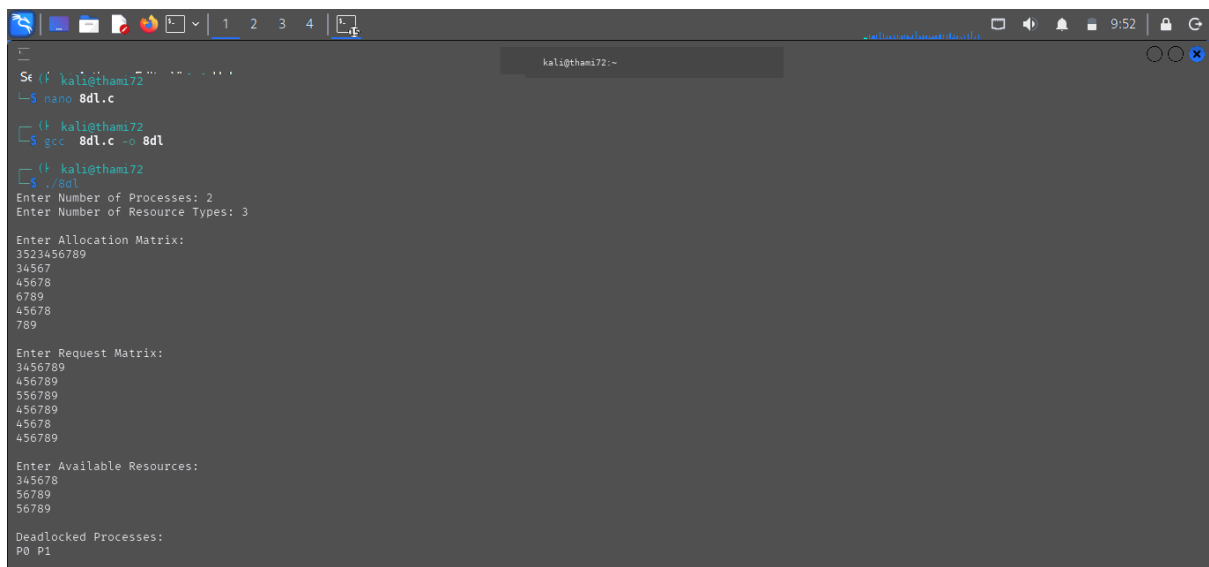


C PROGRAMMING :



```
Se (f kali@thami72 ~)
GNU nano 9.0 8dl.c *
#include <stdio.h>
int main(){
    int allocation[10][10], request[10][10], available[10], work[10], finish[10], n, m, i, j, k, found;
    printf("Enter Number of Processes: ");scanf("%d", &n);
    printf("Enter Number of Resource Types: ");scanf("%d", &m);
    printf("\nEnter Allocation Matrix:\n");
    for (i = 0; i < n; i++){
        for (j = 0; j < m; j++){
            scanf("%d", &allocation[i][j]);}printf("\nEnter Request Matrix:\n");
        for (i = 0; i < n; i++){
            for (j = 0; j < m; j++){
                scanf("%d", &request[i][j]);}printf("\nEnter Available Resources:\n");
            for (i = 0; i < m; i++){
                scanf("%d", &available[i]);work[i] = available[i];
                for (i = 0; i < n; i++){finish[i] = 0;
                    do{found = 0;
                        for (i = 0; i < n; i++){
                            if (finish[i] == 0){
                                for (j = 0; j < m; j++){
                                    if (request[i][j] > work[j])break;
                                    if (j == m){
                                        for (k = 0; k < m; k++){work[k] += allocation[i][k];}finish[i] = 1;found = 1;}}
                                } while (found);found = 0;printf("\nDeadlocked Processes:\n");
                                for (i = 0; i < n; i++){
                                    if (finish[i] == 0){printf("P%d ", i);found = 1;}}
                                if (found == 0){
                                    printf("No Deadlock Detected");}printf("\n");
                                return 0;
                            }
```

OUTPUT :



```
Se (f kali@thami72 ~)
kali@thami72:~$ nano 8dl.c
kali@thami72:~$ gcc 8dl.c -o 8dl
kali@thami72:~$ ./8dl
Enter Number of Processes: 2
Enter Number of Resource Types: 3

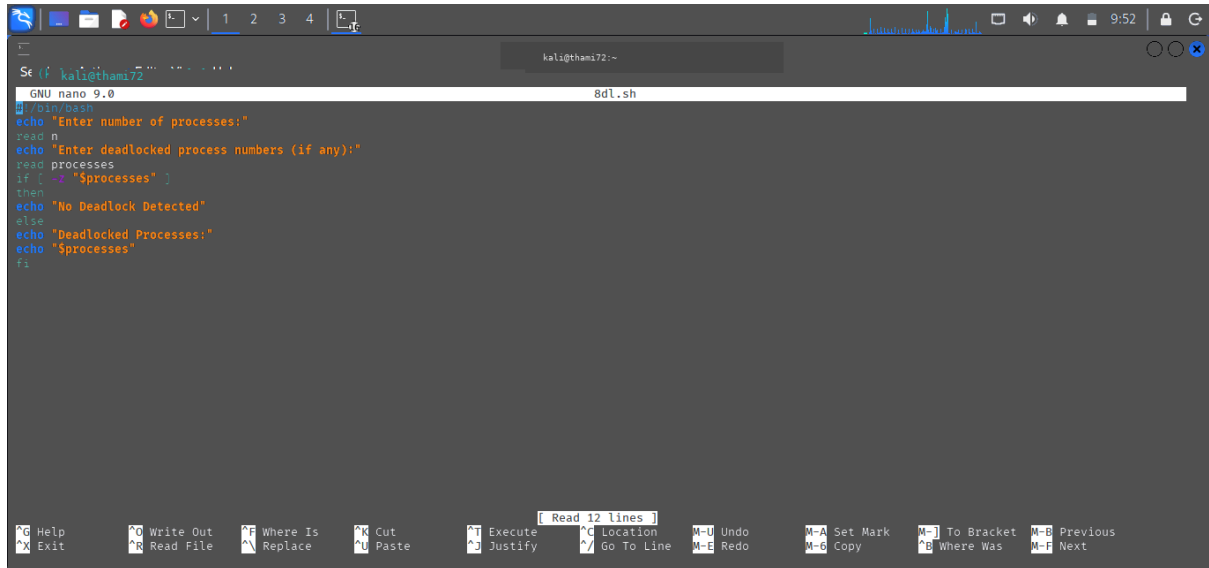
Enter Allocation Matrix:
3523456789
34567
45678
6789
45678
789

Enter Request Matrix:
3456789
456789
56789
456789
45678
456789

Enter Available Resources:
345678
56789
56789

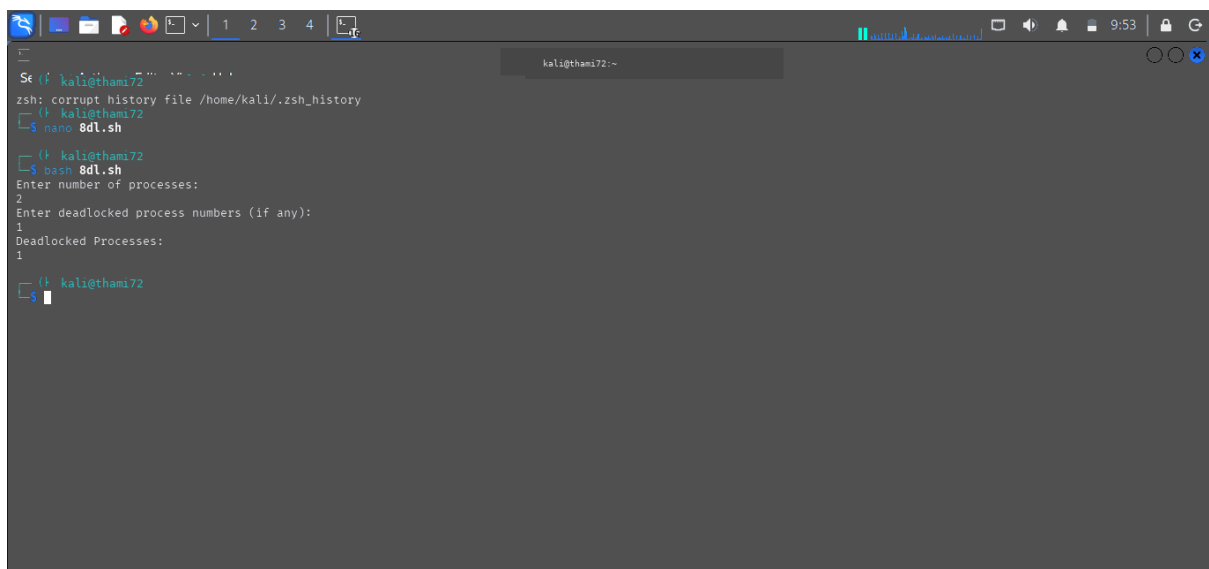
Deadlocked Processes:
P0 P1
```

SHELL PROGRAMMINNG :



```
GNU nano 9.0 8dl.sh
#!/bin/bash
echo "Enter number of processes:"
read n
echo "Enter deadlocked process numbers (if any):"
read processes
if [ -z "$processes" ]
then
echo "No Deadlock Detected"
else
echo "Deadlocked Processes!"
echo "$processes"
fi
```

OUTPUT :



```
zsh: corrupt history file /home/kali/.zsh_history
kali@thami72 ~
$ nano 8dl.sh
kali@thami72 ~
$ bash 8dl.sh
Enter number of processes:
2
Enter deadlocked process numbers (if any):
1
Deadlocked Processes:
1
kali@thami72 ~
$
```